



PL - SERIES

PROFESSIONAL POWER AMPLIFIER

USER MANUAL

Models: PL 4.1 · PL 4.5 · PL 1004 · PL 1504

TABLE OF CONTENTS

§	Section	Page
1	Introduction & Overview	3
2	Product Models	3
3	Package Contents	4
4	Safety Instructions	4
5	Panel Overview — PL 4.1 / PL 4.5	5
6	Panel Overview — PL 1004 / PL 1504	6
7	Technical Specifications	7
8	Connection & Wiring Guide	8
9	DSP & Ethernet Control	11
10	Operating Instructions	12
11	Protection Features	13
12	Maintenance & Troubleshooting	14
13	Warranty & Support	15

1. INTRODUCTION & OVERVIEW

The **PL Series** from Louis Martin Audio represents the pinnacle of professional Class-D power amplifier technology. Engineered for demanding live sound, fixed installation, and touring applications, the PL Series delivers exceptional power output with outstanding audio fidelity while maintaining a compact 2U form factor.

The DSP variants (PL 4.1 and PL 4.5) integrate a **64-bit multimode DSP** with full Ethernet remote control, offering limiters, precision delay, configurable crossovers, and factory speaker presets. The analogue variants (PL 1004 and PL 1504) provide high-performance amplification with comprehensive onboard protection circuitry, making them equally suited to any professional installation.

Key application areas include: concert halls and live performance venues, entertainment lounges, pubs and bars, auditorium and house-of-worship installations, and broadcast or studio monitor systems.

2. PRODUCT MODELS

The PL Series is available in four models across two sub-families:

Model	Sub-Family	Power @8Ω / ch	Power @4Ω / ch	Bridged @8Ω	DSP / Control
PL 4.1	DSP	1000 W × 4	1600 W × 4	3000 W × 2	64-bit + Ethernet
PL 4.5	DSP	1500 W × 4	2500 W × 4	5000 W × 2	64-bit + Ethernet
PL 1004	Analogue	1000 W × 4	1600 W × 4	3000 W × 2	Onboard protection
PL 1504	Analogue	1500 W × 4	2500 W × 4	5000 W × 2	Onboard protection

3. PACKAGE CONTENTS

Please verify all items are present upon opening the packaging. Contact your dealer immediately if any item is missing or damaged.

- 1 × PL Series Power Amplifier unit
- 1 × IEC AC Power Cable (country-specific)
- 4 × NL4 Speakon output connectors (pre-installed)
- 4 × XLR Input connectors (pre-installed)
- 1 × Ethernet Cable (Cat 5e, 1 m) — DSP models only
- 1 × Quick-Start Guide
- 1 × Warranty Registration Card
- 2 × Rack mounting ears and screws

■ *Retain all packaging materials in case the unit needs to be returned for service.*

4. SAFETY INSTRUCTIONS

Please read all instructions carefully before operating this equipment.

- **Do not use this product near water or in wet/damp environments.**
- **Do not expose the unit to rain, moisture, or dripping liquids.**
- Clean only with a soft dry cloth. Never use solvents or abrasive cleaners.
- Do not install near heat sources such as radiators, stoves, or other heat-producing amplifiers.
- Do not block ventilation openings. Always allow at least 1U (44 mm) of free air space above and below the unit in rack installations.
- Route power cables away from audio signal cables to prevent hum or interference.
- Protect the power cord from being walked on or pinched at plug points.
- **Unplug the unit during lightning storms or when left unused for extended periods.**
- **Never remove the top cover while the unit is powered. No user-serviceable parts inside.**
- Refer all servicing to qualified service personnel. Servicing is required when the unit has been damaged or used in smoky/dusty environments.
- Connect only to a properly earthed (grounded) AC supply matching the voltage specified on the rear panel.

5. PANEL OVERVIEW — PL 4.1 / PL 4.5 (DSP Models)

Front Panel:



Fig. 1 — PL 4.5 Front Panel

Item	Control / Indicator	Function
1	LCD Display	Backlit display showing channel status, DSP parameters, input/output levels, IP address and system information
2	Navigation Keys (UP / DOWN / MENU / EXIT / ADJUST)	Navigate the onscreen DSP menu; adjust parameter values
3	ONLINE Button	Connects unit to network / activates Ethernet remote control mode
4	BRIDGE Indicators	Illuminate when channels are configured in bridge (mono) mode
5	Signal / Limit / Clip LEDs	Per-channel metering: GREEN = signal present; YELLOW = limit active; RED = clipping
6	MUTE Button	Mutes all outputs simultaneously; LED confirms mute state
7	Power Switch	Illuminated rocker switch; allow 2-second warm-up after power-on

Rear Panel:



Fig. 2 — PL 4.5 Rear Panel (Inputs / Outputs / Ethernet)

Item	Connector / Feature	Description
A	XLR Input 1–4 (LINK IN)	Balanced XLR line-level inputs; 20 k Ω impedance; 0.775 V / 1.44 V sensitivity
B	XLR Link Out 1–4	Buffered XLR loop-through outputs for daisy-chaining to additional amplifiers or processors
C	Speakon NL4 Output 1–4	Locking Neutrik Speakon outputs; Pin 1+/1– = full-range or high; Pin 2+/2– = low (biamp mode)
D	Ethernet Port	RJ-45 Ethernet for network remote control via PC software; supports static IP and DHCP
E	Mains IEC Inlet	100–230 V AC / 50–60 Hz auto-ranging power inlet with fuse holder

6. PANEL OVERVIEW — PL 1004 / PL 1504 (Analogue Models)

Front Panel:



Fig. 3 — PL 1004 Front Panel

Item	Control / Indicator	Function
1	PROTECT LED (per channel)	Illuminates RED when protection circuitry has engaged (DC, thermal, short circuit, etc.)
2	ACL (Auto Clip Limiter) LED	Illuminates AMBER when the auto clip limiter is actively reducing gain
3	SIGNAL LED	Illuminates GREEN when audio signal is present above -40 dBu
4	RESET Button	Manually resets protection circuitry after fault condition has cleared
5	POWER Switch & LED	Mains power rocker switch with blue illuminated indicator

Rear Panel:



Fig. 4 — PL 1004 Rear Panel

Item	Connector / Feature	Description
A	XLR Input 1–4 (LINK IN)	Balanced XLR line-level inputs; 20 k Ω impedance; default gain 32 dB
B	XLR Link Out 1–4	Buffered XLR loop-through outputs for signal distribution
C	Speakon NL4 Output 1–4	Locking Neutrik Speakon speaker outputs
D	DIP Switch Array	Configures bridge mode and input sensitivity per channel pair
E	Mains IEC Inlet	160–270 V AC / 50–60 Hz; fuse holder adjacent to IEC inlet

7. TECHNICAL SPECIFICATIONS

7.1 DSP Models — PL 4.1 / PL 4.5

Parameter	PL 4.1	PL 4.5
Power @ 8 Ω per channel	1000 W $\times$ 4	1500 W $\times$ 4
Power @ 4 Ω per channel	1600 W $\times$ 4	2500 W $\times$ 4
Bridge @ 8 Ω	3000 W $\times$ 2	5000 W $\times$ 2
Average Current Draw	6.5 A @ 230 V AC	7.5 A @ 230 V AC
Input Power Consumption	1495 W	1725 W
Digital Signal Processor	64-bit multimode DSP with Ethernet remote control	64-bit multimode DSP with Ethernet remote control
Frequency Response	20 Hz – 20 kHz ($\pm$ 0.5 dB)	20 Hz – 20 kHz ($\pm$ 0.5 dB)
THD+N (8 Ω , 1 kHz)	< 0.05%	< 0.05%
S/N Ratio	> 98 dB (A-weighted)	> 98 dB (A-weighted)
Damping Factor (1 kHz @ 8 Ω)	> 1000	> 1000
Crosstalk (1 kHz)	> 85 dB	> 85 dB
Input Sensitivity	0.775 V / 1.44 V	0.775 V / 1.44 V
Input Impedance	20 k Ω Balanced	20 k Ω Balanced
Dimensions (W $\times$ D $\times$ H)	482 $\times$ 330 $\times$ 88 mm (2U)	482 $\times$ 330 $\times$ 88 mm (2U)
Net Weight	10 kg	11 kg

7.2 Analogue Models — PL 1004 / PL 1504

Parameter	PL 1004	PL 1504
Power @ 8 Ω per channel	1000 W $\times$ 4	1500 W $\times$ 4
Power @ 4 Ω per channel	1600 W $\times$ 4	2500 W $\times$ 4
Bridge @ 8 Ω	3000 W $\times$ 2	5000 W $\times$ 2
Average Current Draw	6.5 A @ 230 V AC	7.5 A @ 230 V AC
Input Power Consumption	1495 W	1725 W
Protection Circuitry	Clipping, Long Term RMS, DC, Thermal, Short Circuit, Overloads	Clipping, Long Term RMS, DC, Thermal, Short Circuit, Overloads
Frequency Response	20 Hz – 20 kHz ($\pm$ 0.5 dB)	20 Hz – 20 kHz ($\pm$ 0.5 dB)
THD+N (8 Ω , 1 kHz)	< 0.1%	< 0.1%
S/N Ratio	> 98 dB (A-weighted)	> 98 dB (A-weighted)
Damping Factor (1 kHz @ 8 Ω)	> 3000	> 3000

Parameter	PL 1004	PL 1504
Crosstalk (1 kHz)	> 90 dB	> 90 dB
Input Sensitivity / Default Gain	32 dB	32 dB
Input Impedance	20 kΩ Balanced	20 kΩ Balanced
Dimensions (W × D × H)	482 × 330 × 88 mm (2U)	482 × 330 × 88 mm (2U)
Net Weight	10 kg	11 kg

8. CONNECTION & WIRING GUIDE

This section provides detailed guidance on connecting your PL Series amplifier safely and correctly. Always power down all equipment before making or changing connections.

8.1 System Signal Flow

A typical PL Series installation follows this signal chain:



8.2 Input Connections (XLR)

The PL Series accepts balanced XLR inputs on each channel. Use standard 3-pin XLR cables from your mixing console or processor outputs. The LINK OUT connectors (also XLR) allow you to daisy-chain the signal to additional amplifiers without a Y-split or signal splitter.

XLR Pin	Function	Wiring Note
Pin 1	Ground / Shield	Connect to cable shield / chassis ground
Pin 2	Hot (+)	Positive signal; carry this from console balanced output
Pin 3	Cold (-)	Negative signal; for unbalanced sources, connect Pin 3 to Pin 1

■ For unbalanced (RCA or TS jack) sources, use an XLR adapter and connect Pin 1 and Pin 3 together. Expect a 6 dB reduction in input sensitivity. Keep unbalanced cable runs as short as possible to avoid noise pickup.

8.3 Output Connections (Speakon NL4)

Speaker outputs use locking Neutrik Speakon NL4 connectors. Twist and lock the connector clockwise to secure it. Never unplug Speakon connectors under load — always mute the amplifier first.

NL4 Pin	Stereo Mode	Bi-Amp Mode
1+	CH positive output	High-frequency (HF) positive
1-	CH negative output	High-frequency (HF) negative
2+	Not used (leave open)	Low-frequency (LF) positive
2-	Not used (leave open)	Low-frequency (LF) negative

8.4 Bridge (Mono) Mode Wiring

Bridging combines two channels into a single high-power mono output. This doubles the available voltage swing, delivering approximately double the power into the same load. **Minimum speaker impedance in bridge mode is 8 Ω.**

Step	Action
1	Enable bridge mode via DSP menu (DSP models) or rear-panel DIP switch (analogue models) for the desired channel pair (CH1+CH2 or CH3+CH4)

2	Feed the input signal only to the ODD channel (CH1 or CH3) — the even channel input is internally routed
3	Connect the speaker between Pin 1+ of the ODD channel Speakon and Pin 1+ of the EVEN channel Speakon using an NL4 bridging cable
4	Confirm bridge LED illuminates on the front panel
5	Set source level conservatively — bridged output is significantly more powerful; risk of speaker damage if driven hard into clipping

■ **WARNING: Never connect speakers below 8 Ω in bridge mode. Operating into lower impedances will trigger protection circuitry and may cause permanent damage.**

8.5 Typical Installation Wiring Diagram

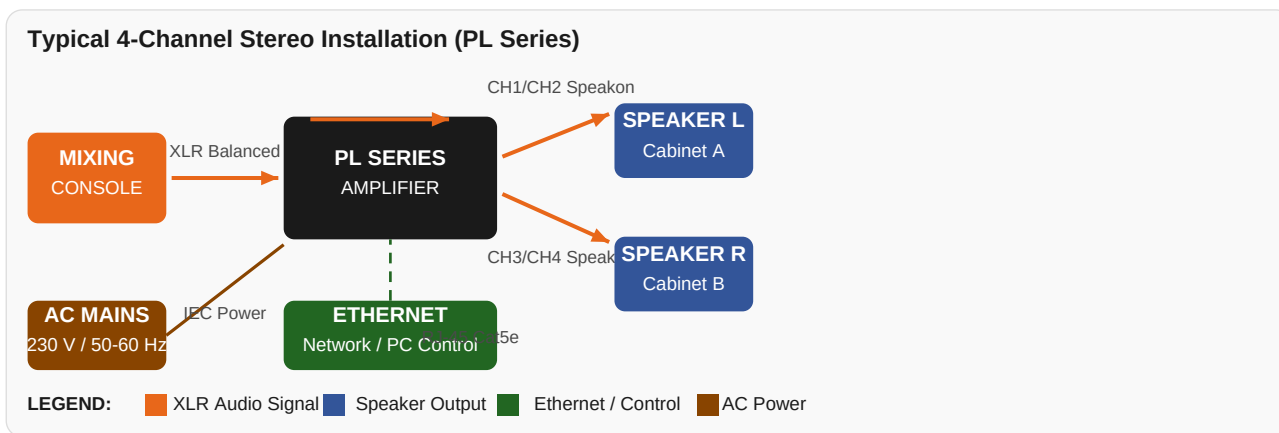


Fig. 5 — Typical 4-Channel Installation Wiring Diagram

9. DSP & ETHERNET CONTROL (PL 4.1 / PL 4.5)

The 64-bit multimode DSP provides comprehensive signal processing on each channel, fully configurable from the front-panel LCD menu or remotely via the Ethernet port using the Louis Martin DSP Controller software (available for Windows and macOS).

9.1 DSP Features

- Input gain control per channel (± 20 dB range)
- High-pass and low-pass crossover filters (Butterworth, Linkwitz-Riley, Bessel — 6 to 48 dB/oct)
- Parametric EQ: up to 8 bands per channel with adjustable frequency, gain and Q
- Delay: up to 200 ms per channel for time alignment in complex speaker arrays
- Limiters: Peak, RMS, and clip limiters individually configurable per channel
- Speaker presets: factory presets for popular cabinet brands; user presets storable
- Input routing matrix: any input can be routed to any output
- Bridge mode configuration via DSP menu

9.2 Ethernet Remote Control

Connect the amplifier to your network using a standard Cat 5e or Cat 6 Ethernet cable. The amplifier supports both DHCP (automatic IP assignment) and static IP addressing.

- **Connect:** Plug a Cat 5e cable between the amplifier's RJ-45 port and your network switch or router
- **Power on:** Power on the amplifier; press ONLINE on the front panel to activate network mode
- **IP Address:** Check the LCD display for the assigned IP address, or navigate MENU > Network > IP Info
- **Software:** Open the Louis Martin DSP Controller software on your PC/Mac and scan for devices
- **Connect:** Select the amplifier from the device list; full parameter control is now available

■ Multiple PL Series amplifiers can be controlled simultaneously from a single PC. Assign unique static IP addresses to each unit to avoid address conflicts on large installations.

10. OPERATING INSTRUCTIONS

10.1 Initial Power-Up Sequence

1. Ensure all audio source equipment volume/gain controls are set to minimum before powering up the amplifier.
2. Connect all signal and speaker cables as described in Section 8.
3. Switch the POWER rocker switch to the ON position. The front-panel LED will illuminate.
4. Wait 2–3 seconds for the internal relay to engage and the DSP to initialise.
5. Slowly increase the source equipment volume to the desired operating level.
6. Monitor the front-panel LEDs: SIGNAL (green) should illuminate; LIMIT (amber) may illuminate briefly at high levels; CLIP/PROTECT (red) should remain off during normal operation.

10.2 Power-Down Sequence

1. Reduce the source volume/gain controls to minimum.
2. Mute the amplifier using the front-panel MUTE button (DSP models).
3. Switch the amplifier POWER rocker switch to OFF.
4. Allow 60 seconds before disconnecting speaker cables (residual charge in capacitors).

10.3 Gain Structure Best Practices

- Set the mixing console master fader at its nominal (0 dB) position before adjusting amplifier gain.
- For the DSP models, use the onboard limiter to protect loudspeakers — set the peak limiter threshold approximately 3 dB below the speaker's rated peak power.
- Use the RMS limiter to protect speakers from sustained high-power programme material (set 6 dB below rated continuous power).
- Crosstalk between channels is better than 85 dB — ensure correct channel routing before the performance.
- For bi-amplified systems, use the crossover DSP function to set the high-pass and low-pass filter frequencies appropriate for your speaker cabinet.

11. PROTECTION FEATURES

The PL Series incorporates comprehensive multi-stage protection to safeguard both the amplifier and connected loudspeakers. All protection functions are automatic and self-resetting (where applicable).

Protection Type	Models	Description & Action
DC Offset Protection	All	Detects DC voltage on the output stage. Disconnects speaker outputs within milliseconds. PROTECT LED illuminates RED. Requires RESET or power cycle.
Thermal Protection	All	Automatic temperature-controlled fan speed adjustment. At critical temperature, output mutes until unit cools. PROTECT LED illuminates. Self-resetting.
Short Circuit Protection	All	Detects speaker output short circuit. Immediately mutes affected channel(s). Self-resetting after fault cleared.
Overload / Overcurrent	All	Monitors output current continuously. Mutes output if sustained overcurrent detected. Check speaker impedance.
Clipping Limiter	All	Automatic Clip Limiter (ACL) reduces gain dynamically when clipping is detected. ACL LED illuminates AMBER (analogue models).
Long Term RMS Limiter	Analogue	Monitors long-term average power. Reduces gain if sustained high-level programme risks speaker damage.
Peak Limiter	DSP	User-configurable threshold; limits peak output level. Prevents instantaneous speaker damage on transients.
Soft Start Relay	All	2-second muted relay at power-on prevents thumps from reaching speakers during initialisation.

12. MAINTENANCE & TROUBLESHOOTING

12.1 Routine Maintenance

- Inspect all cable connections for security every 3 months or before major events.
- Clean ventilation grilles with a soft brush or compressed air (unit powered OFF) every 6 months or when dusty.
- Wipe the chassis with a soft dry cloth only — no solvents.
- Check that rack mounting screws are secure; vibration can loosen them over time.
- Verify that internal fan operates smoothly (you should hear it spin up at moderate volume) during a sound check.

12.2 Troubleshooting Guide

Symptom	Possible Cause	Remedy
No audio output, power LED on	Mute engaged; signal cable fault; protection active	Disengage MUTE; check XLR connections; press RESET or power cycle
PROTECT LED red	DC fault, thermal overload, short circuit	Allow cool-down; check speaker cables for shorts; power cycle; contact service if recurring
Hum / buzz on output	Ground loop; unbalanced cable; cable near power wiring	Use balanced XLR throughout; add ground lift at console; re-route cables
Distortion at low level	Clipping at source; gain structure too high	Reduce source gain; check mixer EQ for peaks; enable limiters on DSP models
Fan noise excessive	High ambient temperature; blocked ventilation	Improve ventilation; add 1U gap in rack; reduce load
Cannot connect via Ethernet	ONLINE not pressed; IP conflict; cable fault	Press ONLINE; check IP settings; try a different cable or port
One channel lower than others	Input gain mismatch; cable fault	Check individual channel gain in DSP menu; swap cables to isolate fault
ACL / Limit LED always on	Input level too high	Reduce source level; increase limiter threshold in DSP

13. WARRANTY & SUPPORT

13.1 Limited Warranty

Louis Martin Audio warrants this product to be free from defects in materials and workmanship for a period of **2 years** from the original date of purchase by the end consumer, when purchased from an authorised Louis Martin Audio dealer.

The warranty covers:

- Manufacturing defects in materials and workmanship
- Component failures under normal operating conditions
- Factory labour for authorised warranty repair

The warranty does not cover:

- Physical damage resulting from misuse, accident, or unauthorised modification
- Damage caused by operating outside the specified voltage or impedance range
- Normal wear and tear (fuses, connectors, cosmetic surfaces)
- Damage caused by failure to follow the safety and operating instructions in this manual
- Units with removed or altered serial number labels

13.2 Technical Support

For technical support, service requests, or to register your product, contact us through the following channels:

Channel	Details
Website	www.louismartinaudio.it
Email	contact@louismartinaudio.it
Warranty Registration	Complete and return the enclosed card within 14 days of purchase
Authorised Service	Contact us by email to locate the nearest authorised service centre

Thank you for choosing Louis Martin Audio. We are committed to providing you with the finest professional audio equipment. Should you have any questions or require assistance, our team is always ready to help.

www.louismartinaudio.it | contact@louismartinaudio.it